

Recall $(a_n)_n \rightarrow l$ if $\forall \varepsilon > 0, \exists N$ s.t. whenever $n > N, |a_n - l| < \varepsilon$.

Ex. let $a_n = \frac{n^2+1}{2n^2}$ $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$

Solu. let $\varepsilon > 0$, want N s.t. $\forall n > N, |a_n - \frac{1}{2}| < \varepsilon$

$$\begin{aligned} |a_n - \frac{1}{2}| &= \left| \frac{n^2+1}{2n^2} - \frac{n^2}{2n^2} \right| \\ &= \left| \frac{1}{2n^2} \right| < \varepsilon \end{aligned}$$

$$\frac{1}{2n^2} < \frac{1}{2n} < \varepsilon$$

$$\Leftrightarrow \frac{1}{\varepsilon} < 2n \Leftrightarrow \frac{1}{2\varepsilon} < n$$

so let N be an integer larger than $\frac{1}{2\varepsilon}$.

let $N \in \mathbb{Z}, N > \frac{1}{2\varepsilon}$.

Then $\forall n > N$ ($n > \frac{1}{2\varepsilon}$).

$$|a_n - \frac{1}{2}| = \left| \frac{1}{2n^2} \right| < \frac{1}{2n} < \varepsilon$$

Thm. let $(a_n), (b_n)$ be sequences.

Assume $\lim_{n \rightarrow \infty} a_n = a, \lim_{n \rightarrow \infty} b_n = b$

Then ① $\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n = a + b$

② $\lim_{n \rightarrow \infty} a_n b_n = ab$

③ If $b_n \neq 0, b \neq 0, \forall n$ large sufficiently

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a}{b}.$$

PF a). let $\varepsilon > 0$ Want N s.t. whenever $n > N$, $|(a_n + b_n) - (a + b)| < \varepsilon$.

Rough work

$$|(a_n + b_n) - (a + b)|$$

$$\leq |a_n - a| + |b_n - b|$$

want $< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$

$$N = \max\{N_1, N_2\}$$

Since $\lim_{n \rightarrow \infty} a_n = a$, $\exists N_1$ s.t. $\forall n > N_1$, $|a_n - a| < \frac{\varepsilon}{2}$

Since $\lim_{n \rightarrow \infty} b_n = b$, $\exists N_2$ s.t. $\forall n > N_2$, $|b_n - b| < \frac{\varepsilon}{2}$

let $N = \max\{N_1, N_2\}$

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| \quad \text{so for } n > N$$
$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$



From functions $f: \mathbb{R} \rightarrow \mathbb{R}$, if $\lim_{x \rightarrow \infty} f(x) = l$, then if we define a seq

$$a_n := f(n)$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = l$$

So if you have a function, then you can construct a sequence + understand its behaviour. What about $\text{seq} \Rightarrow \text{functions}$?

Thm. Sequential Criterion for Limits

Let $f: \mathbb{R} \rightarrow \mathbb{R}$, and $\lim_{x \rightarrow a} f(x) = l$

Then ① If (a_n) is a seq s.t. $\lim_{n \rightarrow \infty} a_n = a$, then $\lim_{n \rightarrow \infty} f(a_n) = l$.

② If $\lim_{n \rightarrow \infty} f(a_n) = l$ for all seqs (a_n) w/ $\lim_{n \rightarrow \infty} a_n = a$, then $\lim_{x \rightarrow a} f(x) = l$.

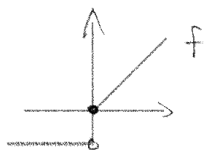
Ex. If $\lim_{n \rightarrow \infty} f(a_n) = l$ for just one seq $(a_n) \rightarrow a$ as $n \rightarrow \infty$

this does not imply that $\lim_{x \rightarrow a} f(x) = l$

$$\text{let } a_n = \frac{1}{n}$$

$$\text{then } \lim_{n \rightarrow \infty} f(a_n) = 0,$$

$$\lim_{x \rightarrow 0} f(x) \text{ DNE.}$$



Pf ① Since $\lim_{x \rightarrow a} f(x) = l$, for any $\varepsilon > 0$

$$\exists \delta > 0 \text{ s.t. whenever } |x - a| < \delta, |f(x) - l| < \varepsilon \quad (*)$$

We're also given a seq $\lim_{n \rightarrow \infty} a_n = a$

Want some N s.t. whenever $n > N$, then $|f(a_n) - l| < \varepsilon$.

$$\text{Since } \lim_{n \rightarrow \infty} a_n = a \quad \exists N \text{ s.t. } \forall n > N, |a_n - a| < \delta$$

$\Rightarrow$ By $(*)$ we know that $|f(a_n) - l| < \varepsilon$.

So we've found the N .

② Suppose $\lim_{x \rightarrow a} f(x) \neq l$ (for a contradiction)

even though $\lim_{n \rightarrow \infty} f(a_n) = l$

$$\forall (a_n) \rightarrow a$$

Goal: Construct (b_n) s.t. $\lim_{n \rightarrow \infty} f(b_n) \neq l$.